

Practice Questions for 14L1-DepthFS

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1 Questions

1. According to the lecture, what is the base case in a recursive tree data structure?
 - A. A node with links to one or more subtrees
 - B. A node with no children
 - C. A null or empty root node
 - D. A node with a value greater than its parent
 - E. A node that has been visited
2. According to the lecture, what is the main difference between Breadth-First and Depth-First tree traversal?
 - A. Breadth-First uses a stack, while Depth-First uses a queue
 - B. Breadth-First visits nodes level by level, while Depth-First traverses down a branch
 - C. Breadth-First is recursive, while Depth-First is iterative
 - D. Breadth-First is used for binary trees, while Depth-First is used for general trees
 - E. Breadth-First is faster, while Depth-First is more memory efficient
3. According to the lecture, what is a key strategy for optimizing recursive algorithms like the Fibonacci sequence calculation?
 - A. Always use iterative methods instead of recursion
 - B. Increase the recursion depth to explore more possibilities
 - C. Remove redundant states and avoid recalculating the same values
 - D. Use a breadth-first search approach
 - E. Use a depth-first search approach

2 Answers

1. According to the lecture, what is the base case in a recursive tree data structure?
 - A. Answer A is incorrect because a node with links to one or more subtrees is the recursive case, not the base case.
 - B. Answer B is incorrect because a node with no children is not necessarily the base case, it could be a leaf node in the tree.
 - C. Answer C is correct because the base case for a recursive tree is a null or empty root node.
 - D. Answer D is incorrect because a node with a value greater than its parent is a property of a binary search tree, not the base case of a recursive tree.
 - E. Answer E is incorrect because a node that has been visited is a concept related to graph traversal, not the base case of a recursive tree.
2. According to the lecture, what is the main difference between Breadth-First and Depth-First tree traversal?
 - A. Answer A is incorrect because Breadth-First uses a queue, while Depth-First uses a stack (implicitly through recursion or explicitly with a stack data structure).
 - B. Answer B is correct because Breadth-First visits nodes level by level, while Depth-First traverses down a branch.
 - C. Answer C is incorrect because both Breadth-First and Depth-First can be implemented recursively or iteratively.
 - D. Answer D is incorrect because both Breadth-First and Depth-First can be used for both binary and general trees.
 - E. Answer E is incorrect because the speed and memory efficiency of Breadth-First and Depth-First depend on the specific tree structure and the problem being solved.
3. According to the lecture, what is a key strategy for optimizing recursive algorithms like the Fibonacci sequence calculation?
 - A. Answer A is incorrect because while iterative methods can be used, recursion can be optimized.
 - B. Answer B is incorrect because increasing recursion depth can lead to stack overflow errors and does not optimize the algorithm.
 - C. Answer C is correct because removing redundant states and avoiding recalculating the same values is a key strategy for optimizing recursive algorithms.
 - D. Answer D is incorrect because breadth-first search is a tree traversal method, not a general optimization strategy for recursive algorithms.
 - E. Answer E is incorrect because depth-first search is a tree traversal method, not a general optimization strategy for recursive algorithms.